

(Part 2 - 15 pieces of furniture)

Proyecto 16: Estantería mural decorativa de 3 niveles con marco interior



1. Description of the furniture

Vertical wall shelving with an open metal structure and a rectangular double-frame design. It features three balanced wooden shelves. Perfect for decorating living rooms, offices, or entryways with an industrial and contemporary touch.

2. General measures

- **Total height:** 100 cm
 - **Total width:** 30 cm
 - **Shelf depth:** 18 cm
 - **Height between shelves:** 32 cm (top), 30 cm (middle), 12 cm (bottom approx.)
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3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	20x20 mm	Sheet Metal 18	100 cm	2 pieces (vertical sides)
Metalon square	20x20 mm	Sheet Metal 18	30 cm	3 pieces (top and bottom outer frame)
Metalon square	20x20 mm	Sheet Metal 18	90 cm	2 pieces (inner frame)
Metalon square	20x20 mm	Sheet Metal 18	18 cm	6 pieces (depth of shelves)

Additional material:

- 3 MDF, pine or melamine wood shelves (30 x 18 cm)
- Invisible screws or fasteners for wood
- Matte black anti-corrosive paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding disc)
- Drill with bits for metal and wood
- Hand sander or sandpaper
- Tape measure, level and set square
- Spray paint or brush

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Shelves (x3)	30 x 18

6. Difficulty level

Low – Excellent option to get started in decorative structures.

7. Estimated manufacturing time

3 to 4 hours (includes welding, painting, and wall mounting)

8. Assembly instructions

1. **Cut all the profiles** for the main structure and the inner frame.
 2. **Weld the outer frame** by joining the two 100 cm pieces with 30 cm crosspieces at the top and bottom.
 3. **Build the displaced inner frame**, fixing it to the structure with welding or screws.
 4. **Add the depth supports** (18 cm) where the shelves will be mounted.
 5. **Sand all joints**, clean and apply black anti-corrosion paint.
 6. **Place the wooden shelves** on the supports and secure them with discreet screws from below.
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9. Additional tips

- You can add **back LED lighting** to give it a modern effect.
- Ideal for small plants, books, picture frames or decoration.
- Secure the wall anchor with **metal anchors** if the weight is considerable.

Proyecto 17: Librero industrial tipo zigzag de gran capacidad



Project 17: Large-capacity zigzag industrial bookcase

1. Description of the furniture

Tall and functional industrial-design bookcase with multiple shelves of varying widths arranged in a staggered pattern. Its structure combines straight lines with Z-shaped modules for greater visual dynamism. Ideal for books, decorative objects, or as a room divider in modern living rooms and offices.

2. General measures

- **Total height:** 180 cm
- **Total width:** 100 cm
- **Depth:** 40 cm
- **Shelf thickness:** 1.5 cm

- **Spaces between levels:** Variable (see sections)
- **Estimated height of each module visible in the image:** between 24 and 35 cm

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Square metalon	25x25 mm	Sheet Metal 18	180 cm	2 pieces (vertical sides)
Metalon square	25x25 mm	Sheet Metal 18	40 cm	10–12 pieces (shelf depth)
Square metalon	25x25 mm	Sheet Metal 18	Variable	Multiple cuts for Z-type reinforcements, vertical and horizontal intermediate reinforcements

Additional material:

- 7 wooden or MDF shelves (see details below)
- Invisible screws or fixings
- Matte black anti-corrosive paint

4. Necessary tools

- MIG or inverter welder
 - Angle grinder (cutting disc)
 - Drill with metal/wood bits
 - Set square, level and tape measure
 - Hand sander or sandpaper
 - Spray paint or brush
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5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Level 1 (top)	100 x 40
Level 2	68 x 40
Level 3	43 x 40
Level 4	53 x 40
Level 5	55 x 40
Level 6	42 x 40
Level 7	50 x 40
Level 8	67 x 40
Level 9 (lower)	97 x 40

6. Difficulty level

Advanced – Due to its size, multiple cuts, progressive assembly and alignment of modules.

7. Estimated manufacturing time

8 to 10 hours (framing, painting, and installing all shelves)

8. Assembly instructions

1. **Cut all metal profiles** according to the measurements of the exterior structure and Z-type modules.
2. **Weld the main vertical and horizontal frames** to form the outline of each level.
3. **Assemble the stepped structure** from bottom to top, ensuring levelness perfect.
4. **Fix the depth reinforcements (40 cm)** and the intermediate vertical ones of each compartment.

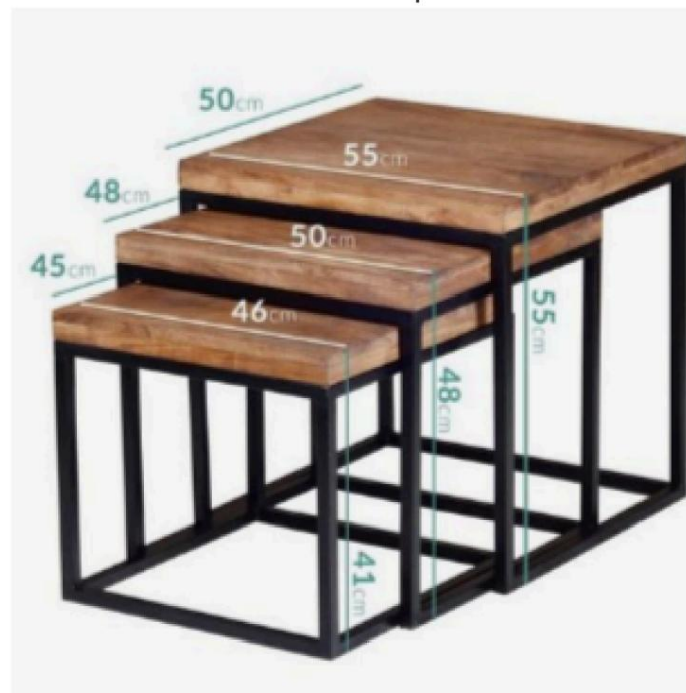
5. **Sand welded joints**, clean and paint matte black.

6. **Place each wooden shelf** on its corresponding level and secure them from below or with invisible supports.

9. Additional tips

- Use a **magnetic square** to maintain exact angles during welding.
- Consider **anchoring it to the wall** if it will be used to store heavy objects.
- You can apply varnish to the shelves for a more rustic or protective finish.

Proyecto 18: Set de mesas auxiliares tipo nido con estructura metálica



Project 18: Nesting Side Table Set with Frame metallic

1. Description of the furniture

Set of three side tables with black metal frames and wooden tops.

Designed to nest inside each other, saving space when not in use. Ideal for living rooms, studios, or as multifunctional side tables.

2. General measures

Large table

- Height: 55 cm
- Width: 55 cm
- Depth: 50 cm

Medium table

- Height: 48 cm
- Width: 50 cm
- Depth: 48 cm

Small table

- Height: 41 cm
- Width: 46 cm
- Depth: 45 cm

3. List of materials (metal structure per table)

Part	Metal profile	Approximate measurements	Amount per table
Metalon base structure 20x20 mm	Vertical sides: according to high		4
Metalon top frame 20x20 mm	Depending on width/depth	4	
Metalon bottom reinforcement 20x20 mm	Optional		1 (recommended)

Additional material:

- 3 solid wood or MDF boards:

- 55 x 50 cm (large table)
 - 50 x 48 cm (medium table)
 - 46 x 45 cm (small table)
 - Screws for wood fixing
 - Black anti-corrosion paint
-

4. Necessary tools

- MIG or inverter welder
 - Angle grinder
 - Drill and bits for metal/wood
 - Sander or sandpaper
 - Square, level, tape measure
 - Paint (spray or brush)
-

5. Cutting details (wooden shelves)

Table Measurement (cm)

Large 55 x 50

Medium 50 x 48

Small 46 x 45
to

6. Difficulty level

Low – Ideal for those who want to make functional furniture with little space and simple precision.

7. Estimated manufacturing time

3 to 5 hours (structure, painting and assembly of the 3 tables)

8. Assembly instructions

1. **Cut the metal profiles** for the three tables according to the measurements.
 2. **Form each rectangular top frame** with four welded profiles.
 3. **Weld the vertical legs** to the top frame of each table.
 4. **Reinforce with lower crossbars if greater strength is desired.**
 5. **Sand all welds**, clean and apply anti-corrosion paint.
 6. **Screw the wooden covers** to the top of each frame.
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9. Additional tips

- Add **rubber or ferrules** to the legs to prevent scratches on the floor.
- You can use **aged boards** for a more rustic style or **light wood** for a Scandinavian look.
- If you want them to fit more easily, **leave a 2 cm tolerance on the frames interiors.**

Proyecto 19: Escritorio industrial con repisas superiores y almacenamiento inferior



Project 19: Industrial desk with top shelves and lower storage

1. Description of the furniture

Work desk with a sturdy metal frame and modern industrial style. It features two levels of upper shelves, a large work surface, and lower compartments for organization. Ideal for offices, home studios, or creative workspaces.

2. General measures

- **Total height:** 175 cm (100 cm upper shelves + 75 cm desk frame)
 - **Total width:** Approximately 120–140 cm (not specified, can be adjusted)
 - **Depth:** 60 cm
 - **Work area height:** 75 cm
 - **Upper shelf heights:** 12 cm (top), 33 cm (middle), 48 cm (bottom)
-

3. Bill of materials (metal structure)

Component	Metal profile	Estimated dimensions	Amount approx.
Vertical structure	Metalon 30x30 mm	175 cm (x2 sides)	2 pieces
Horizontal stringers	Metalon 30x30 mm	According to desk width	4–6 pieces
Metalon 30x30 lower reinforcements		75 cm and 60 cm	4–6 pieces
Sloping sides	Metalon 30x30 mm	85–90 cm (angle cut)	2 pieces

Additional material:

- Work table (e.g. 120 x 60 cm)
- 2 upper shelves (desk width, depth approx. 25–30 cm)
- 1 lower shelf for storage
- Screws and brackets
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
 - Angle grinder
 - Drill with bits for metal and wood
 - Set square, tape measure and level
 - Hand sander or sandpaper
 - Spray paint or brush
-

5. Cutting details (wood surfaces)

Part	Estimated measurement (cm)
Main board	120 x 60
Top shelf	120 x 25
Intermediate shelf	120 x 25
Lower shelf	100 x 25 (adjustable)

6. Difficulty level

Medium – Requires precision leveling, shelf alignment, and stability.

7. Estimated manufacturing time

6 to 8 hours (structure, painting and complete assembly)

8. Assembly instructions

1. **Cut and weld the vertical side frames** (175 cm high), ensuring support for each shelf level.
 2. **Join the sides with horizontal profiles** at the top, middle and bottom.
 3. **Build the base of the desk**, incorporating the crossbars and side reinforcements.
 4. **Sand and paint the metal frame** in matte black.
 5. **Install the desktop board** and then fix the upper and lower shelves with screws or metal brackets.
 6. **Add plastic protectors or levelers** to the legs if they will be placed on delicate surfaces.
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9. Additional tips

- You can add **LED lights** on the upper shelves for decorative lighting or functional.
- For added strength, the main board can be made of **plywood**.
18 mm.
- If mobility is required, you can install **braked casters** on the lower legs.

Proyecto 20: Mueble bajo tipo consola con estantes escalonados



Project 20: Low console cabinet with tiered shelves

1. Description of the furniture

A low, industrial-style console table with a black metal frame and multi-level wooden surfaces. Its tiered design offers aesthetic and organizational functionality for living rooms, entryways, studios, or as a stand for a TV, books, or plants.

2. General measurements (converted from inches to centimeters)

- **Total length:** 59.05 in ÷ **150 cm**

- **Depth:** 15.74 in ÷ **40 cm**
- **Total height:** 26.37 in ÷ **67 cm**
- **Height of lower shelves:** 7.87 in ÷ **20 cm**
- **Height of central module:** 18.89 in ÷ **48 cm**

3. Bill of materials (metal structure)

Component	Metal profile	Estimated measurements	Approx. quantity.
Vertical legs	Metalon 25x25 mm	67 cm	4 pieces
Laterals horizontal	Metalon 25x25 mm	150 cm and 40 cm	6 pieces
Internal structure Z	Metalon 25x25 mm	48 cm, 20 cm, 40 cm	6 pieces approx.

Additional material:

- 3 wooden boards (top, middle and bottom)
 - Top: 150 x 40 cm
 - Middle shelf: 40 x 40 cm
 - Low shelf (left and right): 40 x 40 cm
- Concealed screws and fixings
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder
- Drill and bits for metal and wood

- Set square, tape measure and level
 - Hand sander or sandpaper
 - Spray paint or brush
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5. Cutting details (wooden shelves)

Surface	Measurement (cm)
Top shelf	150 x 40
Stepped middle shelf	40 x 40
Lower shelves	40 x 40 (x2)

6. Difficulty level

Low – Ideal project for functional decorative production with a simple but attractive structure.

7. Estimated manufacturing time

3 to 5 hours (structure, painting, and surface fixing)

8. Assembly instructions

1. **Cut and weld the outer frame** (top and bottom) with 150 cm and 40 cm profiles cm.
2. **Fix the vertical legs** (67 cm) in each corner.
3. **Build the inner Z frames**, ensuring symmetry for the shelves staggered.
4. **Sand the joints, degrease and paint** with a matte black finish.
5. **Install the wood surfaces** with screws or hidden brackets from below.

9. Additional tips

- Add **non-slip rubber** to the base if you will be placing it on delicate floors.
- If the use will be decorative, you can use **recycled or aged wood** for a look more rustic.
- It also works as a low TV stand if reinforced with an extra base.

Proyecto 21: Mesa auxiliar o barra alta con diseño en Z



Project 21: Z-shaped side table or high bar

1. Description of the furniture

Modern industrial design side table with a "Z" shaped frame. This design offers a bold and functional style, perfect for use as a side table, support table, single high bar or even a base for a continuous bar.

2. General measures

- **Total height:** 70 cm (optional also at 45 cm for use as a low table)
- **Board width:** 40 cm
- **Board depth:** 30 cm
- **Inclined structure (Z angle):** approximately 60°–70° relative to the floor

3. Bill of materials (metal structure)

Component	Metal profile	Estimated measurements	Approx. quantity.
Z-shaped structure	Metalon 25x25 mm 70 cm, 45 cm (inclined) 3 cuts		angled
Base support	Metalon 25x25 mm 30 cm (horizontal)		1 piece
Base reinforcement side	Metalon 25x25 mm 15 cm approx.		1 optional piece

Additional material:

- 1 wooden board 40 x 30 cm (2–3 cm thick)
- Invisible screws or fixings
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (angle cutting and roughing)
- Drill with bits for metal and wood
- Set square, level and tape measure

- Hand sander or sandpaper
 - Spray paint or brush
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5. Wooden board details

Surface	Measurement (cm)
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Top board 40 x 30	
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6. Difficulty level

Medium – Precision is required on angle cuts to align the “Z” structure.

7. Estimated manufacturing time

2.5 to 4 hours (frame, paint, and board assembly)

8. Assembly instructions

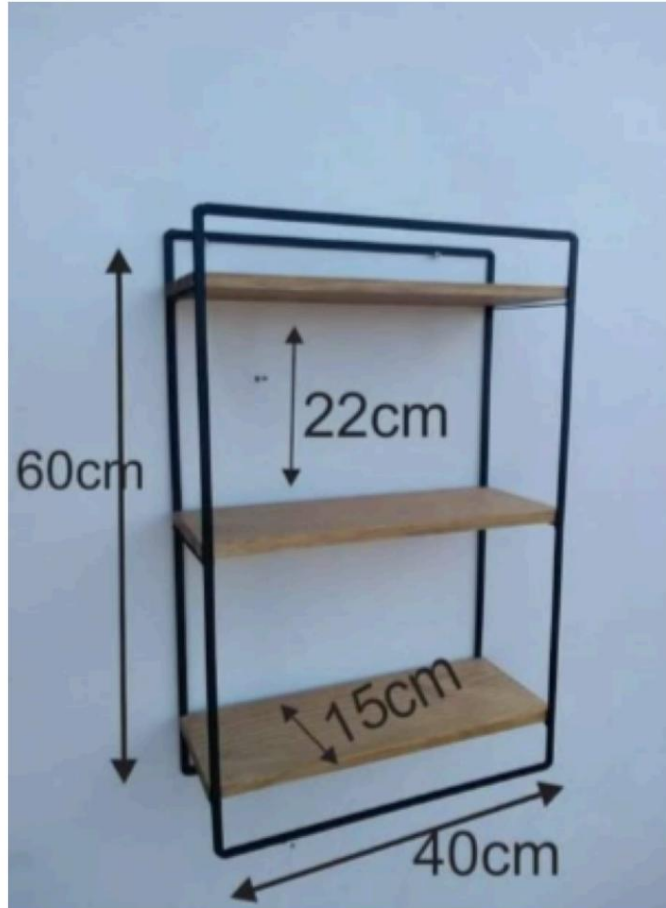
1. **Cut the metal profiles** at the appropriate angles to form the Z.
 2. **Weld the inclined structure**, starting from the base up.
 3. **Add base reinforcement** (30 cm horizontal) for support and stability.
 4. **Place the additional lateral or internal support**, if greater rigidity is desired.
 5. **Sand all joints**, clean and apply matte black anti-corrosion paint.
 6. **Mount the wooden board** on top with screws from below or invisible fixatives.
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9. Additional tips

- If you will be using this design as a bar or coffee station, consider making it **modular** (multiplying the structure in series).

- You can replace the tabletop with **steel, marble or tempered glass** if you want a more sophisticated aesthetics.

Proyecto 22: Estantería mural sencilla de tres niveles



Project 22: Simple three-tier wall shelving

1. Description of the furniture

Minimalist wall-mounted metal shelving with three vertically arranged wooden shelves. Perfect for bathrooms, kitchens, studios, or decorative corners where you want to maximize vertical space without taking up too much space.

2. General measures

- **Total height:** 60 cm

- **Total width:** 40 cm
 - **Depth:** 15 cm
 - **Height between shelves:** 22 cm (approx.)
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3. Bill of materials (metal structure)

Component	Metal profile	Estimated measurements	Amount approx.
Side frames	Round or square 10x10 mm	60 cm	2 pieces
Crossbars horizontal	10x10 mm	40 cm (x3 levels + base)	4 pieces
Internal supports	10x10 mm	15 cm (depth) 6 pieces	approx.

Additional material:

- 3 wooden shelves (40 x 15 cm)
 - Screws for wood fixing
 - Wall plugs and screws to anchor to the wall
 - Black anti-corrosion paint
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4. Necessary tools

- MIG or electric welder (or bolt-on structure)
- Angle grinder
- Drill with bits for metal and wall
- Sander or sandpaper
- Paint (spray or brush)

5. Cutting details (wooden shelves)

Shelves	Measurement (cm)
Top, middle, bottom	40 x 15

6. Difficulty level

Low – Ideal for beginners in blacksmithing or metalworking.

7. Estimated manufacturing time

2 to 3 hours (structure, painting, and wall mounting)

8. Assembly instructions

1. **Cut the metal profiles** according to the size of the frame.
 2. **Assemble the main rectangular frame** with 2 vertical profiles of 60 cm and 2 horizontal profiles of 40 cm.
 3. **Weld the intermediate horizontal supports** at a distance of 22 cm from each other.
 4. **Fix the depth reinforcements (15 cm)** to each shelf level.
 5. **Sand, clean and paint** the entire structure.
 6. **Mount the wooden shelves** on each level and secure them with screws from below.
 7. **Anchor the shelf to the wall** with metal plugs or expansion screws as required.
the type of wall.
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9. Additional tips

- The design can be adapted to a greater height or more levels if desired.

- Ideal for **light decoration**: small plants, jars, picture frames or flavorings.
- Can be left in polished natural steel for a rustic-industrial finish.

Proyecto 23: Banco alto industrial con estructura en X



Project 23: Industrial high bench with X-frame

1. Description of the furniture

High bench with a welded, X-shaped metal frame for maximum strength and visual style. It features a wooden seat and integrated backrest. Its design can support up to 150 kg, making it ideal for modern and functional spaces.

2. General measures

- Total height (to the backrest): 112 cm

- **Seat height:** 82 cm
- **Base width and depth:** 40 x 40 cm
- **Seat:** 38 x 38 cm
- **Backrest height:** 30 cm (from the seat)

3. Bill of materials (metal structure)

Component	Metal profile	Dimensions	Amount approx.
Lower base	Metalon 25x25 mm	40 x 40 cm (square)	4 pieces
Supports vertical/X	Metalon 25x25 mm	82 cm inclined	4 pieces
Seat frame	Metalon 25x25 mm	38 x 38 cm (frame)	4 pieces
Back	Metalon 25x25 mm	30 cm high, 38 cm wide broad	2–3 pieces

Additional material:

- 1 wooden seat 38 x 38 cm (can be individual slats or a single board)
- Invisible screws or fasteners
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding)
- Drill with bits for metal and wood
- Set square, level and tape measure

- Hand sander or sandpaper
- Paint (spray or brush)

5. Cutting details (wood)

Surface	Measurement (cm)
Seat	38 x 38
(Optional) 6 slats of	38 x 6 approx.

6. Difficulty level

Medium/Advanced – X-rigging requires precision for stability and visual symmetry.

7. Estimated manufacturing time

4 to 6 hours (frame, painting, and assembly of seat and backrest)

8. Assembly instructions

1. **Cut the profiles** for the lower and upper frame of the seat.
2. **Assemble the lower square base** (40x40 cm) and the seat frame (38x38 cm).
3. **Weld the pieces in an X shape** , joining the base with the seat and fixing the crossing precisely.
4. **Add the central and backrest reinforcement crossbars.**
5. **Sand all joints**, clean and apply matte black anti-corrosion paint.
6. **Secure the seat wood** with screws from below or invisible fasteners.

9. Additional tips

- You can use **coarse sandpaper and matte varnish** to give the wood a rustic touch.
- For added security, reinforce the X joints with **flat interior plates**.
- This design can also be adapted as **a backless bench**, reducing the total height 82 cm.

Proyecto 24: Librero industrial con compartimentos y gabinete inferior



Project 24: Industrial bookcase with compartments and cabinet lower

1. Description of the furniture

Industrial-design vertical bookcase that combines open and closed storage. Its irregular upper compartments make it ideal for displaying decor, wines, or books, while the lower cabinet with wooden doors offers hidden storage space. Ideal for living rooms, offices, or dining rooms.

2. General measures

- **Total height:** 180 cm

- **Total width:** 90 cm
- **Depth:** 40 cm
- **Bottom cabinet height:** 50cm
- **Upper compartment height (left):** 53 cm
- **Internal compartments:**
 - Width: 26 to 55 cm
 - Height: 22 to 27 cm
 - Door width: 41 cm each

3. Bill of materials (metal structure)

Component	Metal profile	Dimensions	Amount approx.
Structure general	Metalon 25x25 mm	Height, width, depth	10–12 pieces
Internal divisions	Metalon 25x25 mm	22, 26, 27, 55 cm depending on the module	10 pieces approx.
Base reinforcement	Metalon 25x25 mm	Cabinet perimeter	4 pieces

Additional material:

- 4 wooden boards:
 - Superior (90 x 40 cm)
 - Interiors (custom-made to the compartments)
 - Cabinet (2 doors 41 x 50 cm)
 - Hinges, handles and screws
 - Black anti-corrosion paint
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4. Necessary tools

- MIG or inverter welder
 - Angle grinder
 - Drill with bits for metal and wood
 - Sander or sandpaper
 - Set square, tape measure and level
 - Paint gun or brush
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5. Cutting details (wood)

Component	Measurement (cm)
Top shelf	90 x 40
Internal shelves	26, 41, 55 x 40 (depending on module)
Gates of the cabinet	41 x 50 (x2)

6. Difficulty level

Advanced – Requires structural accuracy, door alignment, and multiple compartments.

7. Estimated manufacturing time

10 to 14 hours (complete structure, doors, painting and assembly)

8. Assembly instructions

1. **Cut and assemble the outer frame** 180 cm high, 90 cm wide and 40 cm deep.

2. **Add internal metal partitions** to form stepped compartments.
 3. **Weld the base of the lower cabinet** and reinforce with cross support.
 4. **Fix the wooden boards on the upper levels.**
 5. **Cut and assemble the lower doors** with hinges and handles.
 6. **Sand, clean and paint the entire structure** with an anti-corrosive finish.
 7. **Mount interior shelves** and secure the furniture to the wall if maximum storage is desired.
stability.
-

9. Additional tips

- You can use **parota wood, rustic pine or black melamine** depending on the style desired.
- Ideal to adapt as **a bar cabinet, book display case or entertainment center.**
entertainment.
- Doors can incorporate **metal grilles or glass** if you are looking for an aesthetic different.

Proyecto 25: Soporte vertical para macetas o estantería decorativa



Project 25: Vertical Planter Stand or Decorative Shelf

1. Description of the furniture

Minimalist vertical shelving unit with five tiered levels, ideal for displaying potted plants, books, or decorative items. Its tall, narrow design allows it to fit into tight spaces, such as corners, hallways, or balconies.

2. General measures

- **Total height:** 180 cm
- **Base:** 30 x 30 cm
- **Height between levels:** 42 cm
- **Shelf depth:** 18 to 30 cm
- **Height of the first level:** 20 cm
- **Height of each side column:** 180 cm

3. Bill of materials (metal structure)

Component	Metal profile	Dimensions Dear	Approx. quantity.
Side columns	Metalon 20x20 mm	180 cm	2 pieces
Lower base	Metalon 20x20 mm	30 x 30 cm (square)	4 pieces
Metalon shelf supports	20x20 mm	18 to 30 cm (depending on level)	10–12 pieces
Reinforcement diagonals	Optional	30 cm (x2 lower)	2 pieces

Additional material:

- 5 wooden shelves (30 x 18 to 30 cm)
- Invisible screws or fixings
- Black or white anticorrosive paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder
- Drill with bits for metal and wood
- Hand sander or sandpaper
- Set square and tape measure
- Spray paint or brush

5. Cutting details (wooden shelves)

Level	Measurement (cm)
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Shelf 1 (bottom) 30 x 30

Shelf 2 30 x 28

Shelf 3 30 x 26

Shelf 4 30 x 24

Shelf 5 30 x 18
(top)

6. Difficulty level

Low – Ideal for beginners and very functional for small spaces.

7. Estimated manufacturing time

2.5 to 4 hours (structure, painting and assembly)

8. Assembly instructions

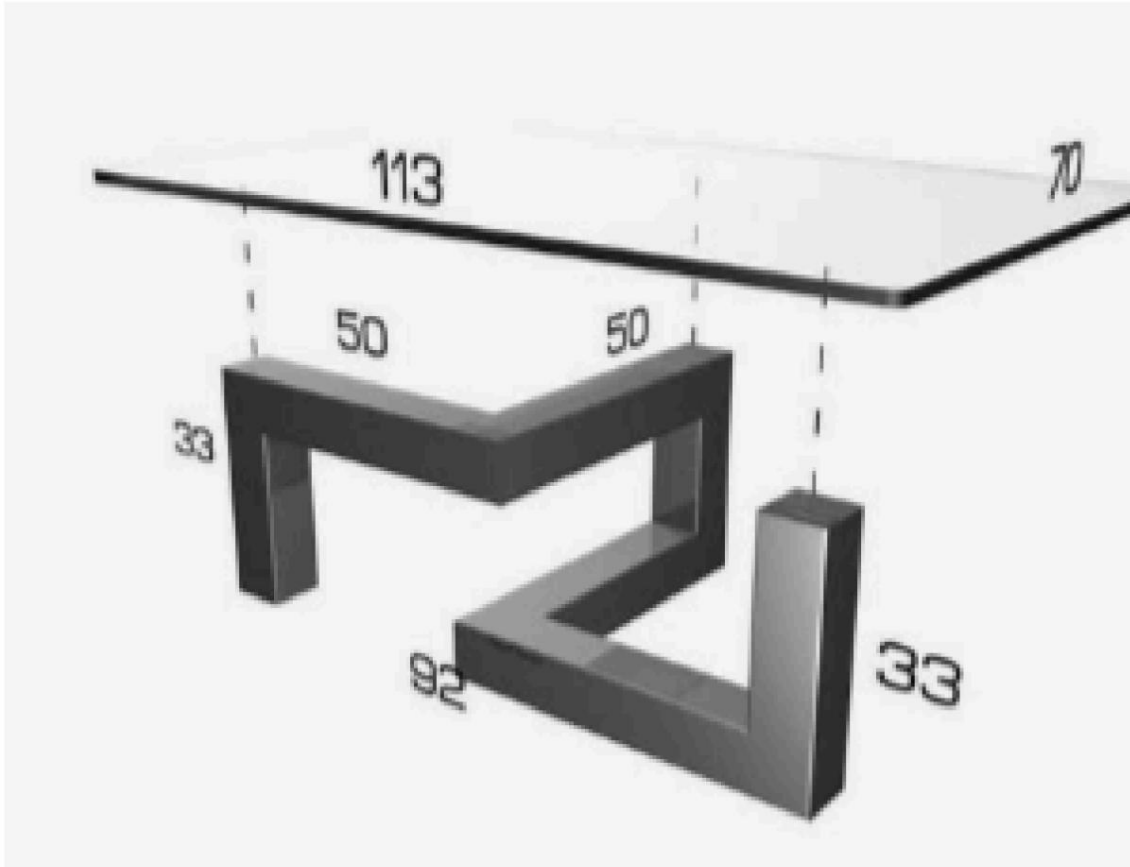
1. **Cut the columns and shelves** to the indicated measurements.
 2. **Form the vertical frame** with the two 180 cm side columns attached to the lower base.
 3. **Weld the horizontal supports** at each height to hold the shelves (every 42 (approx. cm).
 4. **Add lower diagonal braces** for added stability if needed.
 5. **Sand the joints, paint and let dry.**
 6. **Mount the wooden shelves** on the metal supports and secure them from below.
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9. Additional tips

- Excellent option for **covered exteriors or bright interiors.**
- You can varnish the wood with light or intense tones according to your style.

- It can also be adapted as a **bookshelf or vertical picture holder**.

Proyecto 26: Mesa con base geométrica tipo Z continua



Project 26: Table with continuous Z-type geometric base

1. Description of the furniture

Contemporary design table with a continuous Z-shaped metal base, ideal for modern dining rooms, executive desks, or boardrooms. Its striking and structurally sound base combines aesthetic balance and functionality.

2. General measures

- **Top cover:** 113 cm (length) x 70 cm (depth)
- **Total height:** 66 cm (33 cm on each side)

- **Z-type base:** 92 cm (total length)
 - Horizontal segments: 50 cm each
 - Center segment: 92 cm long
 - Width of the profiles: suggested 6x6 cm or 8x4 cm
-

3. Bill of materials (metal structure)

Component	Metal profile	Dimension is	Amount approx.
Lower section	Rectangular or square metal (60x40 mm or 40x40 mm)	92 cm	1 piece
Vertical sections	Same profile	33 cm	2 pieces
Horizontal sections superiors	Same profile	50 cm	2 pieces

Additional material:

- 1 wooden, MDF, tempered glass or marble top: 113 x 70 cm
 - Screws for fixing or hidden anchors
 - Black paint or anti-corrosive metallic varnish
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4. Necessary tools

- MIG or inverter welder
- Angle grinder (with cutting and grinding disc)
- Drill with metal bits
- Hand sander or sandpaper
- Tape measure and set square
- Spray paint or brush

5. Cutting details

Part	Length (cm)
Lower horizontal	92
Verticals	33 (x2)
Superior horizontals	50 (x2)

6. Difficulty level

Medium – Requires precision for square cuts and clean welds at visible angles.

7. Estimated manufacturing time

4 to 6 hours (structure, painting, and roof fixing)

8. Assembly instructions

1. **Cut the metal profiles** to size: 2 vertical, 2 upper, 1 long lower.
 2. **Weld the frame in a continuous Z shape**, starting from the horizontal base, the vertical supports and then the upper ones.
 3. **Sand the welded joints and clean the surface** with degreaser.
 4. **Apply anti-corrosion paint** in matte black or metallic finish.
 5. **Secure the top cover** (wood, glass or marble) with invisible supports, suction cups, anchors or screws depending on the material.
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9. Additional tips

- For a minimalist look, pair with **a tempered glass cover**.

- For a warmer style, use **solid wood in a natural tone**.
- This design can be adapted to **TV tables, desks, bars, or coffee tables**, varying the proportions.

Proyecto 27: Mesa auxiliar en C con ruedas



Project 27: C-shaped side table with wheels

1. Description of the furniture

“C” type side table with metal frame and wheels, designed to slide easily under sofas, armchairs or beds. Its shape allows you to place a laptop, food or books at the reach without taking up much space. Perfect for apartments, offices or work areas rest.

2. General measures

- **Total height:** 60 cm
- **Board width:** 50 cm
- **Depth:** 35 cm

3. Bill of materials (metal structure)

Component	Metal profile	Dimensions	Approx. quantity.
Laterals vertical	Metalon 20x20 mm 60 cm		2 pieces
Lower base	Metalon 20x20 mm 50 x 35 cm (rectangle)		4 pieces
Upper support	Metalon 20x20 mm 50 and 35 cm (frame superior)		2 pieces

Additional material:

- Wooden board: 50 x 35 cm (thickness 1.5–2 cm)
 - 4 swivel wheels (2 with brakes recommended)
 - Screws for fixing the cover
 - Black anti-corrosion paint
-

4. Necessary tools

- MIG or inverter welder
 - Angle grinder (cutting and grinding)
 - Drill with bits for metal and wood
 - Sander or sandpaper
 - Tape measure, set square and level
 - Spray paint or brush
-

5. Cutting details (structure)

Part	Measurement (cm)
Vertical sides 60 (x2)	
Lower base	50 x 35
Upper supports	50 and 35

6. Difficulty level

Low – Quick and easy to manufacture, ideal as a first mobile project.

7. Estimated manufacturing time

2 to 3.5 hours (frame, paint, and wheel/board assembly)

8. Assembly instructions

1. **Cut the metal profiles** according to the measurements of the lower frame and legs vertical.
 2. **Weld the lower structure in a rectangular shape** (50 x 35 cm).
 3. **Attach the vertical legs** to one of the long sides of the rectangle.
 4. **Join the legs to the top frame**, creating the “C” shape.
 5. **Sand and paint the entire metal structure** in anti-corrosive black.
 6. **Screw the wooden lid** onto the top.
 7. **Place the 4 wheels** on the metal base (preferably with brakes on at least two).
-

9. Additional tips

- It can be used as **a work table, mobile tray or laptop desk**.
- For added safety, install **non-slip rubber stops** on the top base if the board is removable.
- It can also serve as **a rolling nightstand** for small rooms.

Proyecto 28: Mesa tipo Z minimalista



Project 28: Minimalist Z-shaped table

1. Description of the furniture

Modern side table with a Z-shaped design, ideal for living rooms, offices, or small spaces. Its angled metal frame allows it to be partially inserted under armchairs or beds for greater comfort and functionality.

2. General measures

- **Total height:** 62 cm
- **Board length:** 45 cm
- **Board depth:** 32 cm
- **Board thickness:** 3.5 cm
- **Bottom base:** 32 x 45 cm (rectangular)

3. Bill of materials (metal structure)

Component	Metal profile	Measures	Approx. quantity.
Lower base	Metalon 20x20 mm 32 x 45 cm (rectangle) 4 pieces		
Supports inclined	Metalon 20x20 mm 62 cm (angled)		2 pieces
Top frame	Metalon 20x20 mm 45 and 32 cm		2 pieces

Additional material:

- Wooden board: 45 x 32 x 3.5 cm
- Invisible screws or fixings
- Matte black or anticorrosive paint

4. Necessary tools

- MIG or inverter welder
 - Angle grinder (cutting and grinding)
 - Drill with bits for metal and wood
 - Set square, level and tape measure
 - Paint (spray or brush)
 - Hand sander or sandpaper
-

5. Cutting details (structure)

Part	Measurement (cm)
Sloping sides	62 (x2)
Lower base	32 x 45
Top frame	32 x 45

6. Difficulty level

Low – Ideal project for beginners with basic experience in welding and assembly.

7. Estimated manufacturing time

2.5 to 4 hours (frame, painting, and board fixing)

8. Assembly instructions

1. **Cut the metal profiles** to form the rectangular bottom base.
 2. **Weld the inclined supports** that will join the base to the upper frame.
 3. **Assemble the top frame** (32 x 45 cm) to receive the lid.
 4. **Sand the joints, clean and apply black anti-corrosion paint.**
 5. **Screw the wooden board** onto the top frame or fix it with adhesive structural.
-

9. Additional tips

- This design can also be adapted as **a nightstand, laptop stand, or breakfast table.**
- You can use **aged or recycled wood** for a rustic touch.
- If you want more stability, add **rubber feet or stops** to the base.

Proyecto 29: Mesa auxiliar tipo C alta y rectangular



Project 29: Tall, rectangular C-shaped side table

1. Description of the furniture

Side or auxiliary table with a "C" frame, specially designed to slide under sofas, chairs, or beds. Its extended height of 80 cm allows greater comfort for users who need to work, eat, or place objects at a higher height.

2. General measures

- Total height: 80 cm
- Board width: 60 cm

- **Board depth:** 40 cm
- **Bottom base:** 60 x 40 cm (rectangular)

3. Bill of materials (metal structure)

Component	Metal profile	Measures	Amount approx.
Vertical legs	Metalon 25x25 mm	80 cm	2 pieces
Lower base	Metalon 25x25 mm	60 x 40 cm (frame rectangular)	4 pieces
Top frame (optional)	Metalon 25x25 mm	60 x 40 cm	2 pieces

Additional material:

- 1 wooden board (60 x 40 x 2–3.5 cm)
- Screws or structural adhesive
- Anti-corrosive black paint or protective varnish

4. Necessary tools

- MIG or inverter welder
- Angle grinder
- Drill with metal/wood bits
- Hand sander or sandpaper

- Set square, level and tape measure
- Spray paint or brush

5. Cutting details (structure)

Part	Measurement (cm)
Laterals vertical	80 (x2)
Lower base	60 x 40
Top frame	60 x 40 (optional if board is supported without frame)

6. Difficulty level

Low – Ideal for fast-paced production, functional projects, or commercial furniture.

7. Estimated manufacturing time

2.5 to 4 hours (frame, paint, and board assembly)

8. Assembly instructions

1. **Cut the metal profiles** for the base, vertical legs and top frame if desired.
2. **Weld the 60 x 40 cm rectangular base.**
3. **Attach the 80cm vertical legs** to one of the long sides of the base.

4. **Weld the top frame** or simply screw the board directly to the paws.
 5. **Sand joints, clean and apply anti-corrosion paint.**
 6. **Secure the wooden board** from below with screws or structural adhesive.
-

9. Additional tips

- Perfect for those who need a **higher table than standard side tables.**
- If you want mobility, you can add **small wheels with brakes** in the corners inferior.
- You can customize the board with **recycled wood, melamine, or even light marble.**

Proyecto 30: Set de pedestales metálicos de diferentes alturas



Project 30: Set of metal pedestals of different heights

1. Description of the furniture

Set of three column-type pedestals with a metal structure and wooden surface.
Its simple and elegant design is ideal for displaying plants, sculptures, pots, vases or decorative pieces. They can be used together or separately depending on the space.

2. Overall measurements (per pedestal)

Pedestal I	Height (cm)	Surface (dear)
High	100 cm	30 x 30 cm
Half	75 cm	30 x 30 cm
Low	50 cm	30 x 30 cm

3. Bill of Materials (Metal Structure per Pedestal)

Component	Metal profile	Dimensions by pedestal	Approx. quantity.
Vertical legs	Metalon 20x20 mm	50, 75 or 100 cm	4 per pedestal
Base frames and superior	Metalon 20x20 mm	30 x 30 cm (square) 8 pieces per	pedestal

Additional material:

- 3 wooden boards measuring 30 x 30 cm (one per pedestal)
- Concealed screws or fixings

- Anti-corrosive black paint or varnish

4. Necessary tools

- MIG or inverter welder
- Angle grinder
- Drill with bits for metal and wood
- Set square, tape measure and level
- Hand sander or sandpaper
- Paint (spray or brush)

5. Cutting details (pedestal structure)

Part	Measurements (cm)
Vertical legs	50, 75 or 100 (x4)
Base and top frame	30 x 30 cm

6. Difficulty level

Low – Highly recommended for beginners, mass production or trade show.

7. Estimated manufacturing time

1.5 to 2 hours per pedestal (structure, painting, and assembly)

8. Assembly instructions

1. **Cut the metal pieces** for the frames and legs of each pedestal.
 2. **Weld the base frame and the 30 x 30 cm top frame.**
 3. **Attach the vertical legs** to the corners of both frames.
 4. **Sand, degrease and paint the metal structure.**
 5. **Mount the wooden board** on top with screws from below.
-

9. Additional tips

- You can use different heights or colors of wood to provide variety.
- They are ideal for **events, shops, galleries or product showcases.**
- If you will use them with heavy plants, **reinforce the bottom base** with a cross or frame solid.